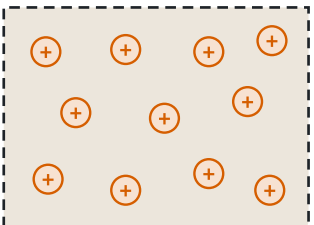
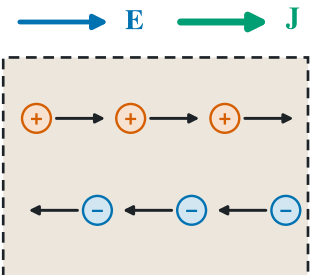
(a) 有净电荷，无电流



$\rho > 0, \quad \mathbf{J} = 0$

例：静电平衡时导体表面的电荷

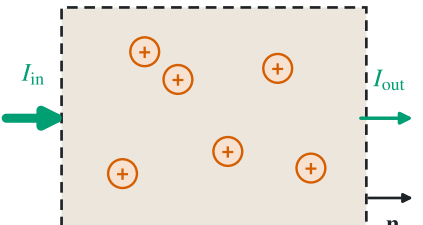
(b) 有电流，无净电荷



$\rho \approx 0, \quad \mathbf{J} \neq 0$

两种载流子对 J 的贡献同向

(c) 流入多于流出



$\frac{dQ}{dt} = I_{\text{in}} - I_{\text{out}} > 0$

虚线是记账边界，不是墙